

MODULE 1 — Tenant & License Management

Document Type: Core Platform SRS

System: Omni-Channel CRM (AI –Led)

Audience: Product, Backend, DevOps, Billing, Compliance, QA

Status: Frozen Architecture Assumptions Applied

1. Module Overview & Business Objective

1.1 Purpose

This module enables a **license-driven, multi-tenant SaaS architecture** where each customer (tenant) operates in complete isolation, and **all platform access is strictly governed by license state and AI usage consumption**.

This module is the **commercial backbone** of the platform:

- Every call, feature, and workflow ultimately resolves to license validity.
 - Revenue leakage prevention is a non-negotiable goal.
 - Compliance, auditability, and enterprise readiness are first-class citizens.
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2. Architectural Assumptions (Frozen & Non-Negotiable)

2.1 Tenant Isolation Model

- **Dedicated database per tenant**
- No shared schema
- No `tenant_id`-based row isolation
- **One database = One Entity/ Company**
- **Max company has multiple database - because data is slow (max 10-15 GB) Logs**

Rationale

- Eliminates cross-tenant data bleed risk
- Simplifies compliance (SOC2, ISO, HIPAA, GDPR -adjacent use cases)
- Nepali date , Audit trail, no direct deletion
- Strengthens enterprise sales posture

2.2 Platform-Owned Telephony

- Platform owns:
 - Phone numbers
 - Telephony providers
 - Routing logic
- Tenants **cannot** bring external telephony credentials
- All billing flows through platform-controlled metering

Rationale

- Prevents revenue bypass
 - Guarantees consistent QoS
 - Simplifies enforcement and analytics
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2.3 Licensing & Pricing Model

- Usage-based licensing
- Primary billable metric: **Credits**
- License duration:
 - Monthly
 - Yearly
 - Custom (date-based)
- Feature tiers are configurable
- **White-labeling: explicitly excluded**

These assumptions are locked unless explicitly revised via change request.

3. Functional Requirements

3.1 Tenant Lifecycle Management

3.1.1 Tenant Creation

System shall allow Super Admin to create a tenant with the following attributes:

Mandatory Fields

- Company Legal Name
- Registered Address
- City
- Country
- Primary Contact Person Name
- Primary Contact Mobile Number
- Primary Contact Email
- Timezone
- Locale
- Currency

Compliance & KYC Fields

- PAN Number
- PAN Certificate (upload)
- Certificate of Incorporation (upload)

Optional Metadata

- Industry (used only for analytics and reporting)
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3.1.2 Tenant Status Lifecycle

Each tenant shall exist in exactly one of the following states:

Status	Description
Active	Full access as per license
Suspended	Temporarily disabled by admin
Expired	License validity ended
Locked	Compliance or security lock
Soft Deleted	Hidden but retained for recovery

Rules

- Soft-deleted tenants are retained for a configurable retention period
- Hard deletion is a backend-only administrative action
- Status changes must be logged with:

- Timestamp
 - Actor (admin/system)
 - Reason code
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3.2 License Configuration & Management

3.2.1 License Attributes

Each tenant shall have at least one active license record containing:

- License Type: Monthly / Yearly / Custom
- License Validity Start Date
- License Validity End Date
- Total Allowed Credits
- Feature Modules Enabled:
 - AI Calling
 - WhatsApp
 - (Future modules configurable)
- Concurrent Call Limit
- Agent Creation Limit
- User Limit (configurable, pricing TBD)

Datta: Not fully functioning with Link showing details of users..

3.2.2 License Number Generation

- License number must be **auto-generated**
 - Format must be deterministic and unique
 - Logic must support prefix customization (internal naming logic to be finalized separately)
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3.2.3 License State Transitions

License may transition between:

- Assigned
- Renewed
- Suspended
- Terminated
- Expired

Rules

- Admin may choose:
 - Immediate effect
 - Next billing cycle effect
 - All transitions must be auditable
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3.3 Usage Tracking & Metering

3.3.1 Metered Metrics

System shall track the following metrics at **real-time granularity**:

Billable

- Credits (primary revenue driver)

Operational Limits

- Number of AI agents created
- Concurrent calls
- Active campaigns

Analytics-Only (Non-Billable)

- CRM record count
- Number of companies
- Number of users

Datta: Not fully functional/Usage analysis to be modifies

3.3.2 Usage Data Integrity[after implementation]

All usage data must be:

- Immutable
 - Time-stamped
 - Non-editable post-recording
 - Queryable for audit and billing reconciliation
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3.3.3 Tier Pricing Configuration

- Tier definitions configurable by Super Admin
- Tier changes do not retroactively affect past usage
- Future usage follows new tier rules

Datta: subscription plan to be editable..

3.4 License Enforcement Rules

3.4.1 Hard Enforcement (Confirmed Behavior)

When **AI minute balance reaches zero** OR **license expires**:

- Outbound AI calls are blocked
- Active campaigns are auto-paused
- Inbound calls:
 - Routed to system message (configurable)
- Tenant UI access restricted (date expiry case)

No silent overages permitted.

Nothing functional Archive data to be maintained for a month

3.4.2 Notifications & Alerts

Threshold-Based Alerts

- Configurable thresholds (e.g., 80%, 90%)
- Channels:
 - Email
 - WhatsApp
 - In-app notification

Zero Balance Alerts

- Immediate notification
 - Custom credit alert messaging supported
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3.4.3 Grace Usage Policy

- No default grace usage
 - Temporary extension allowed **only** via Super Admin approval
 - All exceptions must be logged with reason and expiry
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3.5 Tenant-Level Configuration Scope

Each tenant admin shall be able to configure:

- Business hours
- Default call timezone
- Notification preferences
- Data retention settings (within compliance constraints)

Changes apply immediately unless specified otherwise.

4. Super Admin Controls

Super Admin shall have the ability to:

- Create, edit, suspend tenants
- Assign, override, or revoke licenses
- View aggregated usage across all tenants
- Force-stop campaigns globally
- Lock tenant accounts for compliance violations
- Login secured via OTP + 2FA

All admin actions must be:

- Logged
 - Time-stamped
 - Actor-attributed
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5. Explicitly In Scope

The following are **intentionally excluded** from this module:

- White-labeling
- Feature-tier upsell via tenant UI

- Per-seat pricing logic

These exclusions are strategic to:

- Reduce system complexity
 - Protect revenue clarity
 - Avoid downstream support chaos
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6. Configuration Modules (To Be Defined Separately)

The following configuration modules will be defined in subsequent SRS documents:

- Pricing Engine Configuration
 - Notification Template Management
 - Credit Top-Up & Wallet System
 - Compliance & Retention Policy Engine
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Notes by Shivam

1. Custom Dashboard
2. Content management

Notes for Datta Sir

1. What would be parameters of search ?
2. Leave analytics for now - it would be more customisable later module
3. Verify user logs
4. Review tenant management
- 5.

Admin Figma review

27/03/26

1. Usage of each tenant [No of hours spent/remaining] → PUT User Based filters
2. Recently onboarded tenants/details. → On the dashboard
3. Recent activity/at user level to have much details/call wise...when clicked on user

4. License management/when clicked on license number should show more details in separate window/pop-up → clickable like tenant ID
5. Auto license deactivation on plan expiry and notification to tenet. → need complete notification list from datta sir
6. Usage analysis/ when clicked on new tenants should have more details/tenant wise usage also to be implemented .. → will be taken in another module
7. Manage tenant menu option / create/delete/modify...etc.. → should be done
8. New tenant search option → search logic will be required from HITECH team
9. Payment gateway no implementation in thecredit card/debit card can be implemented. → Need to decide PG and Payment related discussion would be in another module
10. Content management....customize dashboard with pre defined layouts/designs → Dynamic Dashboard and configurable KPIs required
11. Analytics/Alters to be discussed later[Auto payment Notification , Tech support, Payment notification from Bank]
→ To be done later
12. Settings to be done later
→ settings to be done at the last
13. Usage analytics to be based on reports/filters/plugin & play[drag & drop] → it would come in analytics module
14. More interactive dashboard/Usage analysis/Individual Tenant
→ TO be DONE
15. Admin dashboard to have sales/revenue/business analysis/ Customer ratings/Generated leads[weekly/monthly/filter based]/top products sale / volume→ this would be dynamic as well